Petros Papagiannis

he/him

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Links

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Professional Summary

Motivated and versatile Computer Engineering undergraduate with strong expertise in backend, frontend, and full-stack development. Experienced in designing APIs, building scalable web applications, and crafting user-centered interfaces. Skilled at bridging technical and creative solutions, with a passion for problem-solving, continuous learning, and delivering high-quality software. Expanding into cybersecurity with practical experience using reconnaissance, exploitation, and analysis tooling across Linux environments. Engaged in cybersecurity research involving vulnerability analysis, attack surface evaluation, and security mechanism assessment.

Work Experience

S.M.A.R.T.S. IKE

Part-time, Junior Developer

Jun 2025 - Dec 2025

- Full-Stack Development
  - Backend APIs and server-side logic
  - Frontend integration and responsive UI
  - End-to-end UI/UX design and implementation
- Application Architecture
  - Database design and management
  - API design and integration
  - Performance optimization and scalability
  - Design and management of DAG-based data workflows
  - Data pipelining and ETL processes
- AI Large Language Models
  - Integration of cutting-edge LLMs such as GPT and Gemini
  - Development of context-aware conversational AI systems
  - Fine-tuning and prompt engineering for domain-specific applications
  - Implementation of RAG (Retrieval-Augmented Generation) and NLQ (Natural Language Query) systems
  - Machine Learning forecasting and predictive modeling
- Web Mobile Interfaces
  - User-centric design and usability testing
  - Interactive and dynamic web components
- Collaboration Tools
  - Version control (Git) and CI/CD pipelines
  - Agile workflow and team coordination

Lecturing Experience

Computer Engineering & Informatics Department (CEID) – University of Patras

Teaching Assistant - Differential Equations & Applied Mathematics

Oct 2024 Feb 2025

- Assisted in delivering lectures and facilitating practical problem-solving sessions.
- Provided one-on-one support to students on assignments and projects.
- Contributed to the preparation and grading of coursework and exams.

Lab Instructor - Mathematical Foundations in Computer Science

Oct 2024 Jun 2025

- Designed and conducted laboratory sessions on algorithms, formal logic, and proofs.
- Guided students in applying theoretical concepts to practical exercises.
- Assessed lab submissions and provided detailed feedback for improvement.

Education

Computer Engineering & Informatics Department, University of Patras (Undergraduate)

Patras, Greece Oct 2022 and ongoing

Current Grade: 8.50/10

In-depth study: Artificial Intelligence - Machine Learning

Subsidiary subject: Cyber Security

School Education, High school

Kozani, Greece Sep 2019 – Jun 2022

Degree: Secondary school leaving certificate

Grade: 19.9/20

Elective Courses: Mathematics, Physics, Chemistry

School Education, Middle school

Kozani, Greece Sep 2016 – Jun 2019

Degree: Secondary school leaving certificate

Grade: 20/20

Elective Courses: Mathematics, Physics, Chemistry

Projects

Cybersecurity Survey in Military UAV Systems

CEID 2026

Survey and analytical study of cybersecurity threats in military Unmanned Aerial Vehicle (UAV) systems. Analyzes attack surfaces across communication links, navigation, onboard sensors, and ground control stations, focusing on GPS spoofing, jamming, data-link interception, malware injection, and command hijacking. Assesses existing and emerging countermeasures, including cryptographic mechanisms, secure communication protocols, intrusion detection, redundancy strategies, and resilient control architectures.

Cybersecurity, UAV Systems, Network Security, Threat Modeling, Vulnerability Analysis, Secure Communications, Cryptography

Virtualized Penetration Testing Lab

Personal 2026

End-to-end penetration testing environment built using virtual machines, with an isolated attacker and victim VMs. Utilizes Kali Linux as the attacking platform and intentionally vulnerable targets such as OWASP Broken Web Applications and Kioptrix for realistic exploitation scenarios. Covers the full penetration testing lifecycle, including reconnaissance, vulnerability scanning, exploitation, post-exploitation, and privilege escalation, providing hands-on experience with real-world attack vectors, misconfigurations, and security weaknesses in controlled lab conditions.

Kali Linux, OWASP, Kioptrix, Nmap, Burp Suite, VMware

Multi-Dimensional Data Indexing and Similarity Query Processing

CEID 2026

Design and implementation of multi-dimensional indexing methods for efficient k-dimensional queries (k 5). Evaluates k-d trees, quad trees, range trees, and R-trees for multi-attribute datasets, followed by similarity querying on associated textual attributes using Locality-Sensitive Hashing (LSH). Includes a comparative performance analysis of hybrid schemes (tree structure + LSH) with respect to query efficiency, scalability, and accuracy.

Python, Pandas, Locality-Sensitive Hashing (LSH), Spatial Indexing, Performance Evaluation

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Skills

- **Operating Systems:** Linux (Ubuntu, Debian, Kali), Windows
- **Cybersecurity Tooling:** Nmap, Wireshark, Burp Suite
- **Networking & Protocols:** TCP/IP, DNS, DHCP, VLANs, VPN, TLS/SSL
- **Security Testing & Analysis:** Vulnerability Scanning, Penetration Testing, Threat Modeling
- **Cryptography & Security Foundations:** Hashing, Symmetric/Asymmetric Cryptography, PKI, Digital Certificates
- **Programming & Scripting:** C, C++, Python, Bash, PHP, JavaScript, TypeScript, C#, Java
- **Parallelism & Concurrency:** Processes vs Threads, Multithreading, Synchronization Primitives, Race Conditions, Deadlocks, OpenMP, Shared-Memory Parallelism
- **Systems Programming:** Process Scheduling, Inter-Process Communication (IPC), Memory Management, POSIX Threads, System Calls
- **Backend & Runtime Environments:** Node.js
- **Web Technologies:** HTML5, CSS3, Bootstrap, React, Javascript
- **Databases & Data Stores:** MySQL, SQLite, MongoDB
- **Data Science & Machine Learning:** TensorFlow, PyTorch, scikit-learn, Pandas, NumPy
- **Mathematics for ML & AI:** Linear Algebra, Probability Theory, Statistics, Optimization, Calculus
- **Academic Foundations & Systems Knowledge:** Data Structures & Algorithms, Operating Systems, Computer Networks, Distributed Systems, Software Engineering
- **Big Data & Distributed Computing:** PySpark
- **Platforms, DevOps & Tooling:** Git, GitHub, Docker, CI/CD Pipelines, Visual Studio Code, Jupyter Notebook, Apache Airflow (DAGs)

Projects

Open Courses Aggregator Platform

CEID 2026

Full-stack horizontal repository and aggregation platform for open educational courses, integrating multiple external course repositories through REST APIs. Provides a unified React-based front-end for advanced search, filtering, and exploration of courses, backed by a TypeScript-based backend. Leverages Apache Spark with dedicated PySpark jobs for large-scale data processing and machine learning tasks, including course clustering, usage analytics, and recommendation systems. Designed to support scalable ETL pipelines, interoperable data ingestion, and data-driven insights over heterogeneous educational content.

PySpark, React, Python, Typescript, REST APIs, Machine Learning, MongoDB, JSON, Docker

AI-Driven BI Assistant

SMARTS 2025

Full-stack web chatbot for business intelligence dashboards, built with a Python backend and responsive frontend, powered by Gemini LLM. Designed for supervisors, it uses Natural Language Queries (NLQ) to retrieve and interpret BI data, dynamically generating tables, charts, and visual insights. Features include contextual prompt handling, real-time data interaction, and seamless dashboard integration for an interactive and data-driven supervision experience.

Python, HTML, CSS, Javascript, Docker, Voiceflow, OpenAI API, MySQL, AJAX, Plotly, Gemini LLM

Tourism Sites Chatbot - Local RAG

SMARTS 2025

Full-stack web chatbot for tourism websites, built with a Python backend and responsive frontend, powered by Gemini 3-Flash. Uses a local RAG system with vector embeddings to retrieve relevant information from a curated knowledge base, providing real-time, context-aware answers about tourist attractions, events, travel tips, and sightseeing recommendations. Features include custom prompt engineering, dynamic response handling, and seamless website integration for an engaging tourism experience.

Python, HTML, CSS, Javascript, Docker, Voiceflow, OpenAI API, MySQL, AJAX, Gemini LLM

ML Forecasting Pipeline - Airflow & BigQuery

SMARTS 2025

Full-stack machine learning pipeline for time-series forecasting, built with Python and orchestrated using Airflow DAGs. Retrieves and preprocesses large-scale data from BigQuery, trains predictive models, and generates forecasts with automated scheduling and monitoring. Features include modular pipeline design, real-time model evaluation, and seamless integration with dashboards for actionable business insights.

Python, MySQL, Airflow, BigQuery, Pandas, NumPy, ARIMA, Plotly

Public Services Chatbot

SMARTS 2025

Full-stack conversational AI system for public administration, combining Natural Language Query (NLQ) and Context-Aware Generation (CAG) with the Gemini 2.5 Flash LLM. Includes a complete back office for usage statistics, query analytics, and automated data updates. Delivers accurate, context-rich responses while ensuring the knowledge base stays continuously refreshed for reliable, up-to-date information access.

Python, HTML, CSS, Javascript, Docker, Voiceflow, OpenAI API, MySQL, AJAX, Gemini LLM

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Languages

- Greek (Mothertounge)
- English (C2)
- French (B2)

Certificates and Participations

2016	Mathematics Award "Efklidis"
2023	Participation in Best Patras Bootcamp for Full-Stack Development
2024	Teaching Assistant Differential Equations and Applied Mathematics
2025	Lab Instructor Mathematical Foundations in Computer Science
2025	Reference: Prof. Athanasios Andrikopoulos, CEID, University of Patras
2026	Reference: Prof. Nikolas Sklavos, CEID, University of Patras

Recommendation letters available upon request.

Key Strengths

- Structured Problem Solving
- Cross-Functional Collaboration
- Technical Adaptability
- Project Planning & Execution
- Full-Stack Engineering
- Analytical & Critical Thinking
- Secure Engineering Practices

Hobbies

- Basketball, Gym, Swimming
- Exploring emerging technologies, Freelance IT projects
- Traveling and cultural exchange
- Reading tech blogs and scientific articles

Projects

Tourism Sites Chatbot - Voiceflow

SMARTS 2025

Full-stack web chatbot for tourism websites, built with a Python backend and a responsive frontend, powered by Gemini 3-Flash. Developed using Voiceflow, it leverages AI and conversational design to deliver real-time, context-aware answers about popular tourist attractions, local events, travel tips, and sightseeing recommendations. Features include custom prompt engineering, dynamic response handling, and smooth website integration, offering visitors an engaging and informative tourism experience.

Python, HTML, CSS, Javascript, Docker, Voiceflow, OpenAI API, MySQL, AJAX, Voiceflow

Collaborative Thesis Management System

CEID 2025

A Node.js and MySQL web application designed to streamline the administrative process of university diploma theses. Supports students, professors, and administration roles for managing thesis topics, supervision, committee approvals, progress tracking, and final evaluations. Enables collaborative workflow with topic assignment, multi-member committee agreement, document sharing, presentation scheduling, and grading, all within a unified platform.

HTML, CSS, Javascript, NodeJS, MySQL, Figma

UniBook - Collaborative Note Sharing Platform for Students

CEID 2025

Cross-platform desktop app built with Python to enhance student collaboration. Provides a centralized platform for sharing course notes, discussions, and study organization, supporting both student and admin roles with features like uploads, comments, chat forums, and admin moderation.

Python, QTCreator, Robustness Analysis, Project Management

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